

# MILESTONE 1

Problem analysis

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## ***Travel Planner App***

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# 1. ANALYSIS OF RELEVANT LITERATURE

## 1) Time Pressure, and Behavior Change Techniques Affect Route Journey Planning Decisions

**What is it about:** This study analyzes user behavior during route planning in digital apps. The researchers found a major issue: when users face a high cognitive load or time constraints while searching for tickets, they tend to ignore sustainable multimodal routes. Planning a trip involving multiple modes of transportation requires excessive cognitive effort. The authors conclude that to encourage sustainable travel, the app's interface must minimize cognitive demands on the user during the booking process.

**Consequences for our project:** This article gives us a clear direction for the design. It confirms that our Travel Planner App shouldn't just display "dry" data on CO2 emissions. To ensure users (especially busy freelancers like our Persona 2 - Elena) choose sustainable multimodal routes (train + bus), we must make this choice as simple as possible from a UX perspective. We need to implement "seamless" one-click booking for the entire complex route and use the principle of Progressive Disclosure to avoid overloading the screen, as our competitor Skyscanner does.

[1] (Emma Maier, Lewis Turner-Brown, Andrew Broadbent, and Jonathan Freeman. "Attitudes, Time Pressure, and Behavior Change Techniques Affect Route Journey Planning Decisions", MDPI, Released 5 February 2025 on <https://www.mdpi.com/2071-1050/17/3/1297>)

## 2) The impact of progress indicators and information density on users' temporal perception and user experience in mobile pedestrian navigation applications.

Li and Lin (2024) [2] investigate how visual interface elements, such as progress indicators and information density, influence user experience in mobile pedestrian navigation applications.

The study shows that interface design has a significant impact on how users perceive time and interact with the system. In particular, users preferred simple and clear progress indicators (such as progress bars), as they improve learnability and usability. Additionally, presenting too much information can negatively affect user experience, while simpler interfaces help reduce cognitive load and improve overall satisfaction. These findings are relevant for our travel planner application, as it will need to display complex route information in a clear and understandable way. Therefore, our app will focus on intuitive interface design, minimizing unnecessary information, and using simple visual elements to enhance usability and user experience.

[2] Li, S., & Lin, Z. (2024). *The impact of progress indicators and information density on users' temporal perception and user experience in mobile pedestrian navigation applications*. **Displays**, **82**, 102603. <https://doi.org/10.1016/j.displa.2023.102603>

## 3) Route Planning and Cognitive Load.

This research investigates how users decide how to plan their journey under the influence of time constraints. Users tend to use less complicated options when they are faced with complicated interfaces. As well, multimodal routes impose an additional cognitive load that often makes them less likely to use both multi-modal alternatives and the best route possible. Cognitive load should always be reduced where possible through the use of simple structured layouts and reduced complexity in interaction. Too much information and a lack of structure can lead to confusion for the user and therefore negatively impact their ability to make good decisions. This information is relevant to our Travel Planner application as it will help us design the way we present users with complex routes in a more straightforward manner to enable them to make better informed and informed decisions.

[3] Maier, E., Turner-Brown, L., Broadbent, A., & Freeman, J. (2025). *Attitudes, Time Pressure, and Behavior Change Techniques Affect Route Journey Planning Decisions*. Sustainability, 17(3), 1297. <https://www.mdpi.com/2071-1050/17/3/1297>

#### **4) Enhancing Travel Planning Using a User-Centric Approach**

Nikam et al. (2025) [4] explored the design of a unified travel app to solve the problem of travelers feeling overwhelmed by fragmented digital tools. Their study found that existing applications often lack integrated route planning and real-time personalized suggestions. Furthermore, usability testing revealed that while most users appreciate simple interfaces, older users often struggle and require extra onboarding guidance. For our Travel Planner project, this emphasizes the need to integrate all user preferences (time, price, CO2) into one platform while ensuring an intuitive design with clear instructions to accommodate older users.

[4] Avishkar Nikam, Indresh Kumar Verma, and Pranita Ranade. 2025. An Interactive Mobile Application to Enhance Travel Planning Using a User-Centric Approach: A UX Design Case Study. International Journal of Interactive Mobile Technologies (IJIM), 19(17), 115–128. <https://online-journals.org/index.php/i-jim/article/view/51279>

## 2. ANALYSIS OF COMPETITOR PRODUCTS

### 2.1. Rome2Rio

Rome2Rio is a travel planner that lets users search for ways to travel between two places and offers different ways to get to those places (train/bus/plane/ship/car/other). This is similar to our concept of Travel Planners in that it combines transportation options from multiple modes into a single route, provides an overview of the options available to the user as they plan their trip.

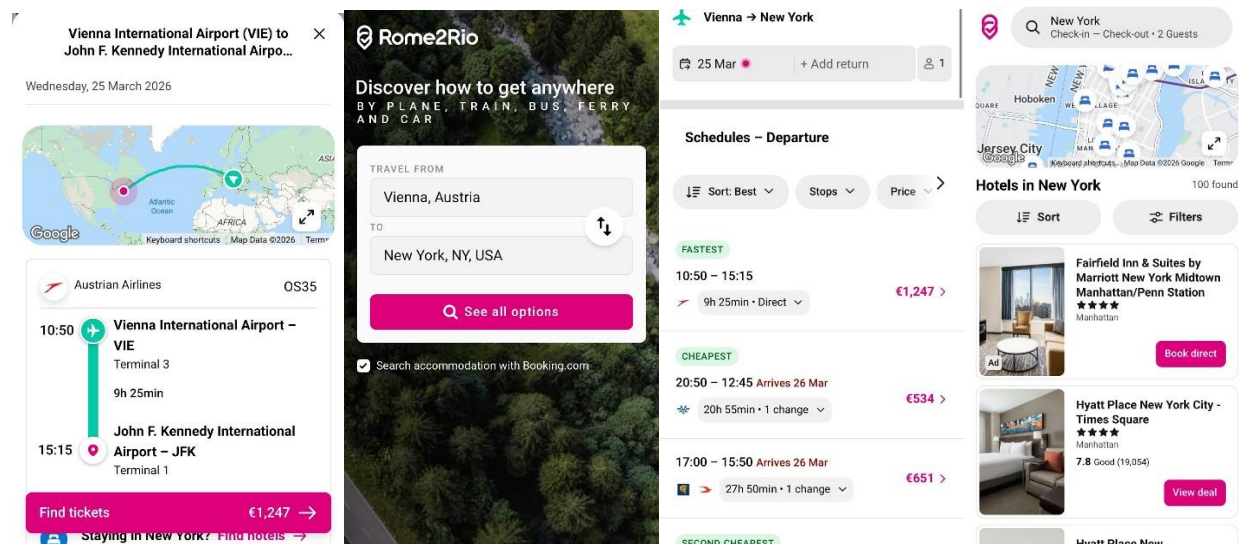
#### Benefits:

- There are many different types (concepts) of transportation available to users.
- There is a clear overview of the route options available to users because each route option is presented very clearly.
- The user interface is relatively straightforward and easy to understand by new users.

#### Limitations and Missing Features:

- Low accuracy: Prices and times of events are often approximate, so they are less reliable.
- No integrated booking: Users are taken to another website to complete their booking and interrupt the flow of the user's experience by not completing the process on the Rome2Rio website.
- Lack of personalization: There are no advanced filters for features such as budget, CO2 impact, and number of transfers.
- Overloading of information: If there are several routes to compare, it will take longer to review them individually because the user will have to compare them all.

Although Rome2Rio illustrates the usefulness of integrating various modes of travel in one platform, it lacks some important elements such as personalization and seamless booking. The Travel Planner App seeks to provide improved information accuracy, improved personalization, and a more fluid user experience.



## 2.2. Skyscanner

Skyscanner is a very big and popular global travel search engine. It focuses on flight comparisons, also offers hotel bookings and car rentals. The system collects data from hundreds of airlines and travel agents to help users find the cheapest, fastest or “best” travel options. For our Travel Planner App, Skyscanner is a major competitor in the field of route planning and budget optimization.

### Pros:

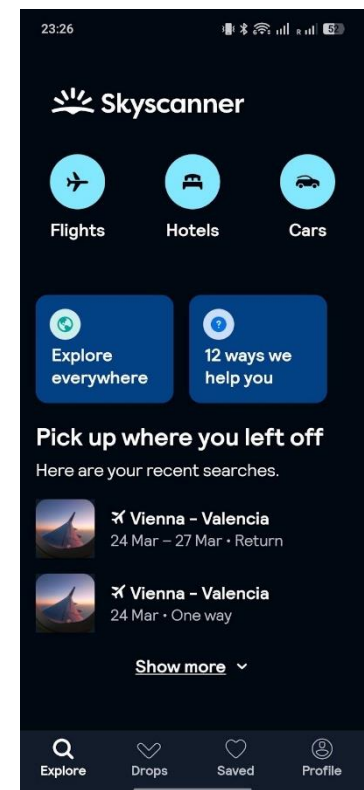
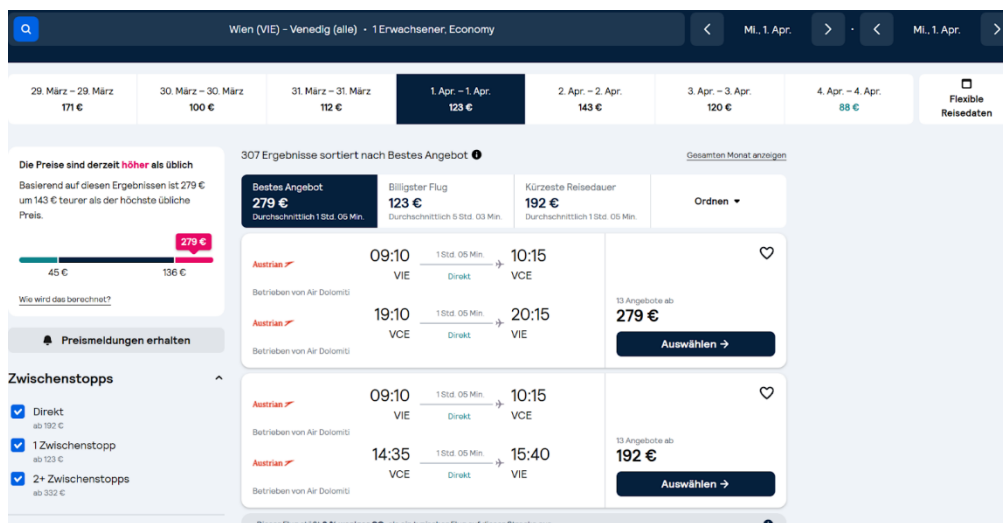
- **Flexible search features:** The “Explore Everywhere” and “All Month” calendar views are great tools for users with flexible dates and destinations. This helps them find the cheapest tickets.
- **Eco-indicators:** Skyscanner highlights flights with lower CO2 emissions with the “Greener choice” label. This feature is ideal for eco-conscious users (like our Persona 1).
- **Price Alerts:** The app offers a robust notification system that tracks price changes for specific routes.

### Cons & Problems:

- **Functionality Disadvantage:** Skyscanner has a big issue, that it focuses only on flights, making it impossible to combine different transportation modes, like plane + train + bus, into a single route from point A to point B. This is a major problem for users who search for easy planning.
- **Booking process:** To actually purchase a ticket, users are redirected to different sites (airlines or agencies). This disrupts the natural user flow, creates unnecessary friction, and sometimes leads to unexpected hidden fees on the partner’s site.
- **Interface Overload (HCI Issue):** The search results page can be visually overloaded with sponsored results and third-party ads.

### Relevance to Our App:

- Skyscanner’s success with flexible date search and CO2 tracking proves that users really value these features. However, its lack of functionality, like the inability to create comprehensive itineraries, highlights exactly what our Travel Planner App needs to fill.



## 2.3. Moovit

Moovit is a public transport navigation app that helps users plan routes using buses, trains, trams, and metro systems. It provides real-time updates and step-by-step navigation for urban travel.

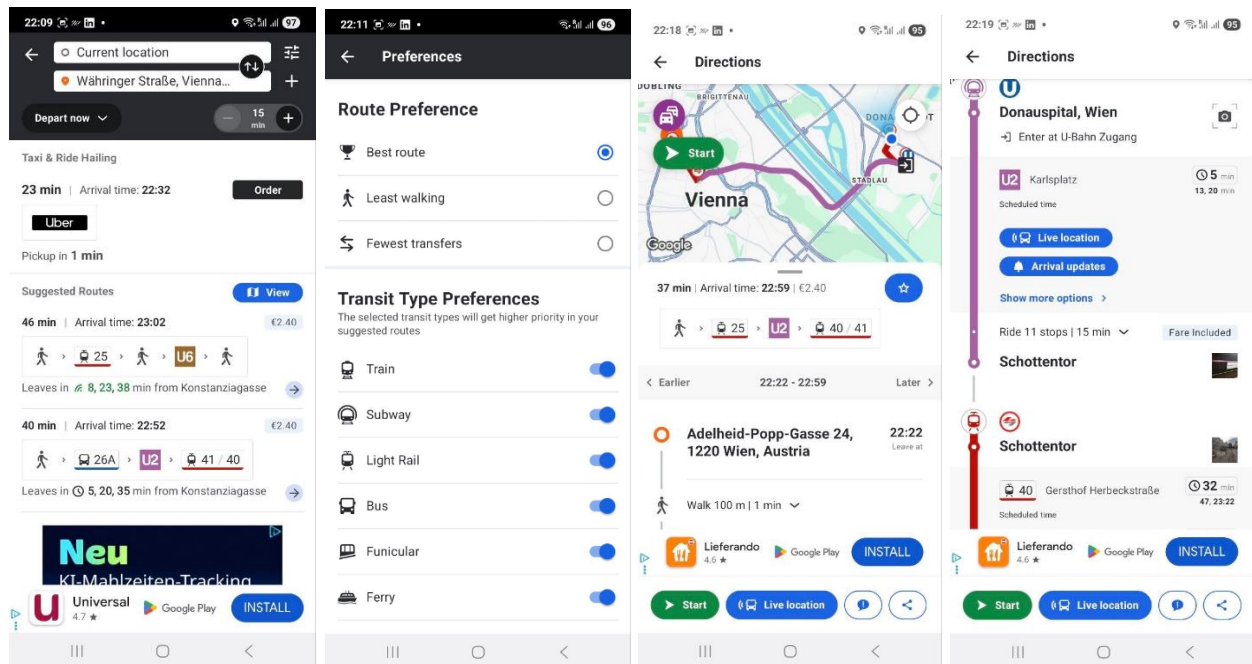
### Advantages

- **Real-time updates:** Provides live information about delays and arrival times
- **Step-by-step navigation:** Guides users during the journey
- **Wide coverage:** Available in many cities worldwide (even smaller ones)
- **User-friendly interface:** Easy to understand routes and schedules

### Disadvantages

- **Limited transport types:** Focuses mainly on public transport and does not include flights or long-distance travel
- **No personalization:** Routes are not adapted to user preferences (e.g., budget, eco-friendliness)
- **Urban focus only:** Not suitable for planning complex or international trips
- **No full trip optimization:** Does not combine different transport types into one optimized journey
- **Intrusive advertisements:** In the free version, frequent ads interrupt the user experience and make navigation less comfortable

Moovit is a great tool to navigate through the city, based on your real time location. However, our application will go beyond this by integrating multiple transportation types (including long distance travels) and providing personalized route recommendations based on user preferences.





## 2.4. Omio

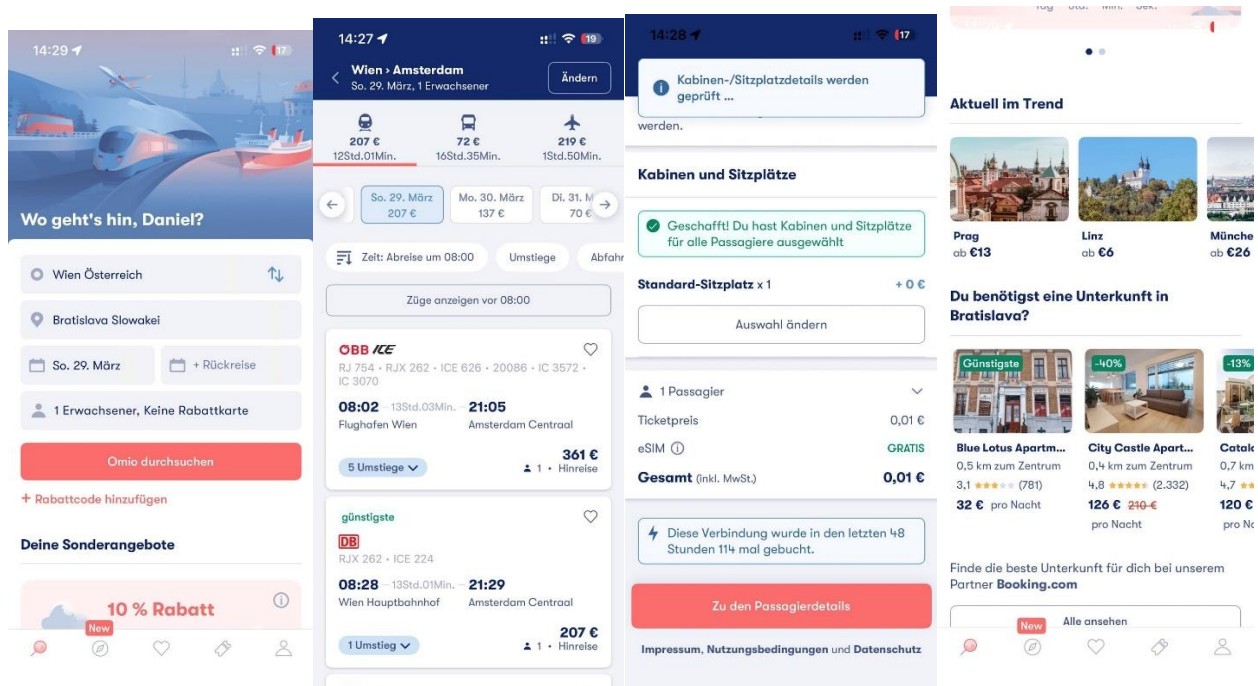
Omio is a travel planning application that integrates various modes of transport: trains, buses, and flights. It closely echoes our "Travel Planner" concept by helping users select routes based on travel time, cost, and transport type to get from one city to another as conveniently as possible.

### Advantages:

- **Quick orientation among options:** Tabs on the results screen allow for an instant comparison of travel times and costs across different transport modes without any extra steps.
- **A guiding interface:** Labels such as "cheapest" and "fastest" (günstigste / schnellste) help the user avoid spending time on detailed analysis, allowing them to select a suitable option more quickly.
- **Easy interaction start:** The search screen is designed to be as clear as possible—large fields for the route, dates, and number of passengers make the start of the search process feel natural and almost intuitive, requiring no explanation.

### Disadvantages and Missing Features:

- **Disregard for environmental factors:** The app focuses solely on price and time, completely omitting information about CO2 emissions and user ecological preferences, which are a vital part of our concept.
- **Limited customization options:** While the user can select a transport type, they cannot set more precise parameters, such as allowable transfer times or comfortable walking distances. There is also no mechanism to account for the user's previous choices.
- **Pressure on the user:** Notifications such as "This route was booked 114 times in last 48 hours" create a sense of urgency and can push users toward hasty decisions, reducing their sense of control.





### 3. USER GROUPS ANALYSIS

Based on the developed personas and the general idea of the Travel Planner app, several major user groups can be determined.

#### **Budget-oriented travelers (students)**

The primary concern of such users is saving costs associated with traveling. They tend to be flexible in terms of time spent traveling but are willing to take a long route or make several transfers if it helps save some money. They may also take into consideration environmental factors when choosing between two options. However, such users face a problem since relevant information can be found on several platforms, which makes planning a time-consuming process.

#### **Time-oriented travelers (freelancers, businessmen)**

For such users, time is a primary concern. They like routes that are fast and efficient, with a minimum number of transfers. They need extremely precise information about the trip since they often have to take some luggage along. They need comfort during the trip as well. Among the main problems faced by such users are unexpected delays, uncomfortable transfers, and the need to purchase tickets from different platforms, which disrupts the planning process.

#### **Casual travelers (experience-oriented users)**

Such users don't want to spend much time in planning their routes. They want simple and clear suggestions rather than studying a variety of options. They might get confused by excessive information and a variety of choices, especially if they have to change their application frequently.

#### **Elderly or less tech-savvy users**

Such users usually want a simple and direct route from A to B and don't want to get into sophisticated filtering and route combination. They want clear interfaces with a limited number of steps to interact with an application. If the interface is too sophisticated or offers too many features, they might abandon this application and opt for simpler solutions or traditional solutions without any application.

#### **Eco-conscious travelers**

Such users are keen to reduce their impact on the environment during their travel. They prefer to travel by train, bus, or other environmentally friendly transportation, unlike flying. Such users need clear information about CO2 emissions during their routes and want to compare routes easily based on their sustainability. Lack of clear information about CO2 emissions and simplicity in comparing routes might discourage them from adopting environmentally friendly routes.

## 4. CONTEXT ANALYSIS

The Travel Planner app will be mainly used by mobile device users, especially smartphone users. The target group will be using the app either immediately before the trip or during the trip. In both situations, the user may be in a rush and may not be able to fully focus on the app.

There are two main situations in which the Travel Planner app can be used. The first one is when planning the trip in advance, most probably at home. In such a case, the user may have enough time to consider all the options. The second case is when the user is already on the way to the desired place, for instance, at a train station or at a bus stop. In such a case, the user needs to be quickly informed about the departure time, possible delays, or platform changes.

Another important aspect to be considered is the environment in which the user is in. The user may be in a crowded place and may be under stress, especially when there is a need to catch a train or a bus. Moreover, the internet access may be unstable, and this may also affect the way the app is being used.

Taking all these aspects into consideration, it can be said that the app needs to be simple and clear. The user needs to be quickly aware of his or her route without having to read too much.

## 5. USER PERSONAS

### Primary Persona 1

**Name:** Albert Wesber

**Age:** 22

**Gender:** Male

**Location:** Vienna, Austria

**Occupation:** Student at TU Vienna

**Motto:**

Explore the world and leave only footprints in the sand, not a carbon footprint.

**Relevant Patterns of Behavior:**

Albert travels through Europe three to four times a year. He tries to avoid flying because of the environmental impact and prefers trains or buses like FlixBus. He is willing to choose longer routes if it is cheaper and more environmentally friendly.

**Social life:**

Travels mainly with a small group of friends from university. They share accommodation costs (staying in hostels or via Airbnb) and plan their itineraries together.

**Goal and needs:**

- Find the cheapest travel options.
- Easily combine different modes of transport (for example, train + bus) to save money.
- View the carbon footprint (CO2 impact) of each route option.

**Pain points:**

- To find a convenient route with transfers, Albert has to open 10 different tabs and websites.
- Trains are often more expensive than budget flights, which goes against his desire to be environmentally friendly.
- The fear of missing a connection during transfers (for example, if the train is delayed and the bus has already left).

**Typical usage scenario:**

Albert and his friends want to go to Italy for the weekend. He opens the app (a travel planner app), enters the destination, and sets the filters: "Budget up to 100 euros," "Minimal CO2 emissions," and "Maximum 2 transfers." The app suggests an ideal route, which he immediately forwards to his friends.

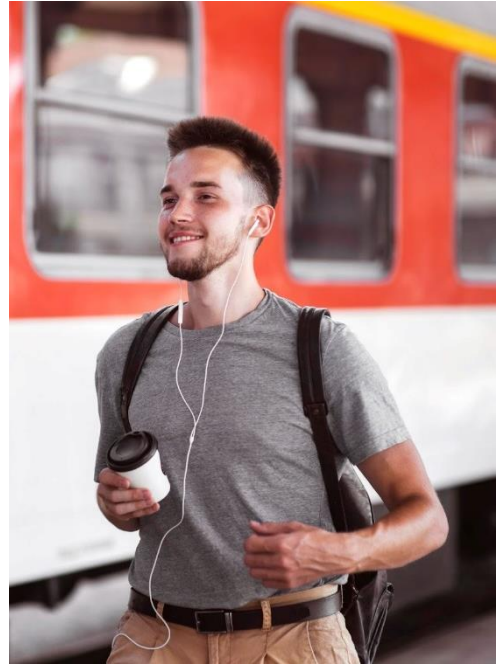


Image Source: [https://www.freepik.com/free-photo/side-view-man-wearing-earphones\\_10422559.htm](https://www.freepik.com/free-photo/side-view-man-wearing-earphones_10422559.htm)

## Primary Persona 2

**Name:** Elena Rossi

**Age:** 28

**Gender:** Female

**Location:** Vienna, Austria

**Occupation:** Freelancer, UX/UI-Designer

**Motto:**



My time is my money. Comfort on the go is more important to me than saving a few euros.

### **Relevant Patterns of Behavior:**

Elena works remotely and often travels from city to city for a few weeks at a time. A stable internet connection and the ability to work while traveling are important to her.

### **Social life:**

She usually travels alone. In new cities, she meets other freelancers at meetups or in coworking spaces.

### **Goal and needs:**

- Optimizing travel time (finding the fastest route from point A to point B).
- Minimizing the number of transfers, as she is traveling with a heavy suitcase and laptop.
- Knowing the exact departure and arrival times to schedule calls with clients.

### **Pain points:**

- She hates unexpected delays or flight cancellations that disrupt her work schedule.
- Difficult transfers with lots of luggage (for example, having to run 15 minutes from the train station to the bus stop).
- There is no combined ticket for different modes of transport (you have to buy everything separately).

### **Typical usage scenario:**

Elena urgently needs to get from Vienna to Munich before her call on Monday morning. She uses the app and prioritizes arrival time and minimum travel time. The app suggests a route that includes everything: a taxi to the train station, a high-speed train, and the subway in Munich – all within a single travel time.

Image Source: [https://www.freepik.com/free-photo/medium-shot-woman-campervan\\_20551599.htm](https://www.freepik.com/free-photo/medium-shot-woman-campervan_20551599.htm)

## Primary Persona 3

**Name:** Camille Dubois

**Age:** 23

**Gender:** Female

**Location:** Paris, France

**Occupation:** Intern in Fashion Marketing

**Motto:** “The journey matters - but not the hassle”

### Relevant Patterns of Behavior

Camille enjoys traveling for short trips, fashion events, and social experiences. She often chooses destinations based on trends or recommendations from social media. She does not like spending too much time planning and prefers simple, ready-made options. Camille usually uses multiple apps but often feels overwhelmed by too many choices.

### Social Life

She is highly active on social media and often shares her travel experiences. She travels with friends or meets people during trips and values aesthetic and enjoyable experiences.



### Goals and Needs

- Plan trips quickly and easily
- Get clear and simple travel options
- Discover new destinations and experiences
- Balance comfort and price without overthinking

### Pain Points

- Too many options and information make planning stressful
- Difficult to compare routes across multiple apps
- Time-consuming decision-making process
- Lack of simple, personalized suggestions

### Typical Usage Scenario

Camille wants to travel from Paris to Milan for a fashion weekend. She opens the app and expects quick suggestions without complex filtering. The app provides a few optimized routes with clear summaries (price, time, comfort level). She quickly selects one and proceeds without needing to compare multiple platforms.

Image Source: [https://www.freepik.com/free-photo/smiley-woman-looking-away\\_9138346.htm#fromView=search&page=1&position=42&uuid=d4397115-303e-40d3-a013-8bfa11342579&query=fashion+intern+travel](https://www.freepik.com/free-photo/smiley-woman-looking-away_9138346.htm#fromView=search&page=1&position=42&uuid=d4397115-303e-40d3-a013-8bfa11342579&query=fashion+intern+travel)

## Secondary Persona



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**Name:** Kalus Ziegler

**Age:** 68

**Gender:** Male

**Location:** Munich, Germany

**Occupation:** Retired Teacher

**Motto:** "Travel should be enjoyable, not stressful."

**Patterns of Behavior:** Klaus visits his family several times a year or goes on short trips with his wife. He knows how to use a smartphone for simple tasks—WhatsApp, calls, or photos—but quickly gets lost in apps with too many buttons and complex navigation.

**Social life:** He frequently spends time with his grandchildren and members of his hiking group. He enjoys discussing trips with his wife, most often over a cup of morning coffee.

**Not very suitable for Travel Planner due to:**

- **Too many filters:** Klaus isn't interested in comparing eco-metrics or complex route customization. He just needs the simplest train.
- **Information overload:** If the app suggests too many mixed transport options (like train + bus + scooter), he gets overwhelmed and just goes to the train station directly to buy a ticket instead.
- **Doesn't need detailed plan:** Getting from Point A to Point B is enough for him. He plans his day in his head, not through an app.

**Typical usage scenario:** Klaus is planning to visit his grandkids in Berlin. He opens the app only to find a direct train route. He has no interest in any additional features. He checks the departure time, buys his ticket, and closes the app until his trip.

## Negative Persona



[https://www.freepik.com/free-photo/male-traveler-making-his-coffee-van\\_18952043.htm](https://www.freepik.com/free-photo/male-traveler-making-his-coffee-van_18952043.htm)

**Name:** Max Spitz

**Age:** 23

**Gender:** Male

**Location:** Berlin, Germany

**Occupation:** Freelance Photographer

**Motto:** "No plans, no problems."

**Patterns of Behavior:** Max loves spontaneous travel. He often travels in his camper van or using hitchhikes. He never knows where he will sleep the next day. He just drives or travels depending on where the weather is good or where he meets new cool people.

**Social life:** Max is very outgoing and meets new people on the road all the time. He hangs out at camper spots and makes decisions randomly most of the times, sometimes based on what his new friends are doing that day.

**Unsuitable for Travel Planner due to:**

- **No fixed routes:** He doesn't take public transport, trains, or flights, so he doesn't need public transport's route suggestions.
- **Hates planning:** The whole point of the app is to plan a trip ahead of time. Max hates planning and prefers to decide what to do in the exact moment.
- **No time limits:** He doesn't care about arrival times or travel duration because he never has a deadline.

**Typical usage scenario:** Max installs the Travel Planner app after hearing about it from someone at a hostel. He opens it, notices that he needs to enter a specific destination and select a time, and closes it right away. He realizes the app is designed for people who prefer a strictly planned vacation. A few minutes later, he deletes it, finding it completely useless for his spontaneous lifestyle.



## 6. TASK ANALYSIS

| User Group                | Task                                     | Importance | Frequency  |
|---------------------------|------------------------------------------|------------|------------|
| Budget-oriented travelers | Find the cheapest available route        | High       | Very often |
| Budget-oriented travelers | Combine different transport types        | High       | Often      |
| Time-oriented travelers   | Find the fastest route                   | High       | Very often |
| Time-oriented travelers   | Minimize number of transfers             | High       | Medium     |
| Casual travelers          | Get simple and quick route suggestions   | Medium     | Often      |
| All users                 | Compare routes (time, price, CO2 impact) | High       | Often      |
| All users                 | Select or book a complete journey        | High       | Medium     |
| Elderly users             | Find a direct and simple connection      | Medium     | Often      |

## 7. PROJECT MANAGEMENT ASPECTS

### Context

Our project focuses on developing a travel planner application that helps users create complete travel routes based on their preferences. Many existing tools require users to compare multiple platforms, which makes the planning process complex and time-consuming. Our application aims to simplify this process by integrating different transportation options and presenting optimized routes in a single interface.

### Motivation

Travel planning can be stressful due to the large number of available options and fragmented information across different platforms. Users often struggle to find routes that balance time, cost, comfort, and environmental impact. The motivation behind this project is to reduce this complexity and provide a user-friendly solution that supports personalized and efficient decision-making.

### Goals

- Provide personalized travel route recommendations
- Integrate multiple transportation types (train, bus, flight, etc.)
- Allow filtering based on user preferences (price, time, CO<sub>2</sub> impact, comfort)
- Reduce planning time and cognitive load
- Improve usability and user experience through intuitive design

### Non-goals

- The app will not replace booking platforms (e.g., ticket purchasing is not the main focus)
- It will not provide real-time tracking of transport (optional future feature)
- It is not intended for professional travel agencies or logistics planning
- It will not include overly complex customization that may overwhelm users

### Team Introduction

We are a team of four students working on a Human-Computer Interaction project focused on designing a travel planner application. Each team member contributes to different aspects of the project, including user research, personas, task analysis, and system design. Through this project, we aim to improve our understanding of user-centered design and develop practical skills in creating intuitive digital products.

### Work Distribution

- **Anastasiia Trofymenko:**  
Project management aspects, 1 primary persona, document integration
- **Artur Ponomarenko:**  
Creation of 2 primary personas

- **Oleh Shynkaryk:**  
User group analysis, context analysis, task analysis
- **Danylo Prokhorenko:**  
Secondary persona, negative persona, competitor analysis
- **All team members:**  
Each contributed one literature analysis and one competitor analysis